

Date	Task	Time	Team	Comments
Oct 11, 2025	Project Kickoff	2.5 hrs	ALL	Finalized tech stack and project phases.
Oct 14, 2025	Frontend Scaffolding	2.0 hrs	SH	Initialized Vite, React, and Tailwind CSS.
Oct 18, 2025	TomTom API Config	2.5 hrs	KP	Registered API keys and tested incident feeds.
Oct 22, 2025	Backend API Init	2.0 hrs	SH, KP	Set up Flask boilerplate and basic routing.
Oct 25, 2025	Dashboard Layout	3.5 hrs	SH	Built the core navigation and sidebar UI.
Nov 1, 2025	Data Parsing Logic	2.5 hrs	KP	Scripted logic to filter TomTom traffic data.
Nov 5, 2025	UI Data Binding	2.0 hrs	SH, KP	Connected TomTom feed to frontend state.
Nov 8, 2025	Map Integration	3.0 hrs	SH	Integrated Google Maps React-Wrapper.
Nov 12, 2025	Incident Overlays	2.5 hrs	SH, JS	Designed CSS for incident pop-up overlays.
Nov 22, 2025	Search Bar Logic	3.0 hrs	SH	Implemented location search and autocomplete.
Nov 29, 2025	Walking Route Logic	3.5 hrs	KP	Initial implementation of TomTom walking API.
Dec 6, 2025	DB & Cloud Config	2.0 hrs	JS	Configured MongoDB and Google Cloud console.
Dec 13, 2025	OAuth Foundations	3.0 hrs	JS	Set up Google Cloud credentials for login.
Dec 16, 2025	Backend Security	2.0 hrs	SH, JS	Implemented JWT for secure API sessions.
Dec 20, 2025	Winter Break: Dijkstra	4.0 hrs	KP	Built Dijkstra's algorithm for transit paths.
Dec 21, 2025	Transit Logic Review	2.5 hrs	SH, KP	Integrated transit math with the frontend.
Dec 23, 2025	Winter Break: OSRM	4.0 hrs	KP	Configured OSRM as the backup routing engine.
Dec 27, 2025	Winter Break: Testing	3.5 hrs	JS	Debugged MongoDB connection strings.
Dec 28,	OSRM Failover	3.0 hrs	KP	Scripted logic for automatic API

2025	Logic			switching.
Jan 10, 2026	MongoDB Integration	3.0 hrs	JS	Linked user profile logic to the DB cluster.
Jan 17, 2026	OAuth Finalization	2.5 hrs	JS	Finalized secure Google login for the app.
Jan 24, 2026	User State Logic	2.0 hrs	SH	Handled persistent login state in React.
Jan 28, 2026	API Performance	2.0 hrs	SH, KP	Refactored Flask routes for faster response.
Jan 31, 2026	AI Model: Logic	3.5 hrs	KP	Wrote the heuristic safety score function.
Feb 7, 2026	Safety Score UI	3.0 hrs	SH	Created color-coded risk levels for the map.
Feb 14, 2026	Route Midpoint Math	3.0 hrs	KP, JS	Logic to calculate safety per segment.
Feb 21, 2026	AI Model Refinement	2.5 hrs	KP	Added location "noise" for safety score variety.
Feb 28, 2026	Socket.IO Init	2.5 hrs	JS	Set up real-time server for live incident data.
Mar 7, 2026	Spring Break: VOSK	4.0 hrs	SH	Integrated VOSK for speech recognition.
Mar 8, 2026	Voice Commands	3.5 hrs		Linked voice input to the routing search.
Mar 10, 2026	Spring Break: Three.js	4.0 hrs	SH	Began 3D walking view environment setup.
Mar 11, 2026	3D Graphics Polish	3.5 hrs	SH	Added simple meshes and lighting for 3D view.
Mar 14, 2026	Spring Break: Performance	4.0 hrs	KP	Optimized Python logic to handle 3D data load.
Mar 15, 2026	Performance Testing	3.0 hrs	SH, KP	Tested UI responsiveness during 3D rendering.
Mar 21, 2026	Real-time Incident UI	3.5 hrs	JS, SH	Synced Socket.IO events to React alerts.
Mar 28, 2026	Code Audit	3.0 hrs	KP	Reviewed safety score logic and TomTom calls.
Mar 31, 2026	CSS Final Polish	3.5 hrs	SH	Cleaned up Tailwind classes for accessibility.
Apr 2, 2026	WPRDC Data Prep	2.5 hrs	JS	Prepared data structures for future crime data.

Apr 4, 2026	QA & Final Debug	4.0 hrs	ALL	End-to-end testing of voice, 3D, and safety.
Apr 6, 2026	Documentation	3.5 hrs	ALL	Finalized README and technical handoff docs.
Apr 8/2026	Hotfixes	3 hrs	ALL	Fixed minor technical issues
Apr 8, 2026	Deployment	4.0 hrs	SH, JS	Deployed final build to production cloud.